

PURCHASE DESCRIPTION
BIG CLIFF POWERHOUSE DRAINAGE PUMP REPLACEMENT PURCHASE
PR 261T021

Scope:

This Purchase Description (PD) covers the purchase of two (2) vertical lineshaft pumps for Big Cliff Powerhouse.

Background:

Big Cliff Powerhouse is a remotely operated site that has been in commission since 1953 both generating power and regulating the flow of the Santiam River. Our facilities have elaborate drainage schemes for various floor drains and sumps which all collect at a common "drainage sump". Under normal operation, the facility has two drainage pumps in lead-lag control to maintain the level of the drainage sump. These pumps have been maintained throughout their service life of 30+ years, but spare components are no longer available for upkeep or refurbishment. Last year, one of the pumps completely seized and the other pump began showing the same signs the failed pump gave before seizing.

Justification:

With only one pump in service, our capacity to drain the facility without intervention is halved. As a temporary measure, we were able to suspend a submersible pump in the drainage sump to support the remaining vertical lineshaft pump.

The addition of this submersible pump allows us to drain the sump should the remaining pump fail but requires an Operator to be sent to site to run them. On initial effort, the crew was unable to tie the submersible pump to the existing lead/lag control system. Due to this limitation, sump level indication will be used as notification to send an Operator to site rather than to turn on the pump.

With Operator availability already being a limiting factor for operations in the valley, we are concerned about the opportunity cost of making on site drainage sump level maintenance at Big Cliff a regular job activity instead of an automated process. This issue is made significantly worse during weekends when only one operator is normally scheduled, meaning under regular circumstances, there is nobody free to go to Big Cliff to operate the submersible pumps.

By replacing these pumps with new commercially available pumps: we will allow us to buy back the reliability of our drainage system, take responsibility for directly maintaining sump levels off the hands of operators, and give us a platform that we should be able to maintain for another 30+ years of operation.

Requirements:

Two (2) Vertical lineshaft pumps

Quantity	Description
2	Outlet: 4" 150# ANSI 16.5 Flange compatible Flow: 75 GPM at 70ft head Static Head: 70 ft Pump Curve: Continuously rising Pump Formfactor: Vertical lineshaft with multistage impellers at inlet Pump Height: 45' from inlet to centerline of outlet flange Lineshaft: 1" diameter, 5' sections, 400 series stainless steel Lineshaft Couplings: Coupling design capacity shall be the published manufacturer service rating divided by 1.5 Lineshaft Bearings: Water lubricated except for motor base bearing which may use oil or grease Impeller Material: Bronze Bowl Material: Cast Iron Prelubrication: If needed, a prelubrication water line will be provided Column Pipe: 5' sections with flanged connections, 400 series stainless steel Pumped Media: Potentially silty water, temperature ranging 39°-65° F Filtering: Inlets must have basket strainers Motors: Up to 3.0 hp with a service factor of 1.15 or greater Motor Power Supply: 3 phase, 480 V, 60 hz circuit Pump Speed: 1800 rpm or less

Required Data:

Shop drawings and any applicable performance data sheets including pump curves. Project to review orders and drawings prior to purchase.

Projected Cost:

Estimated cost is:

Shipping:

Pumps shall be delivered to Detroit Dam, 40990 Highway 22, Mill City, OR, 97360. The Government shall not be responsible for the cost of shipping to Detroit Dam. Deliveries must be made Monday-Thursday between 0700-1600.